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Project Title:	Prototype Testing for Non-Invasive Determination of Intracranial Pressure		
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Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Smart Medical Systems Team		
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Comments:			
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No. of Bachelor's Candidates:	3	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
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Intracranial pressure (ICP) is a vital determinant of brain function whose monitoring and treatment are central to the successful management of patients with a variety of brain diseases and disorders. There are also indications that changes in cerebral autoregulation - one of the primary physiologial mechanisms for maintaining normal ICP, and detectable via alteration of blood flow into the brain - can be altered during space flight in ways associated with space motion sickness and orthostatic intolerance. Currently, ICP is measured by drilling a hole into the skull and placing a device into one of several places within the cranium. Besides the risks attendant with this procedure, measurements of ICP can only be performed in hospitals staffed by neurosurgeons. This is clearly not an option available to astronauts in space, who often develop brain edema and its associated increase in ICP. Therefore, a simple, non-invasive method of monitoring ICP (nICP) would allow measurement of this critical parameter in astronauts. We have developed an algorithm for nICP based on novel analysis of standard diagnostic ultrasound of the brain and invasively and continually monitored arterial

Task Description:	blood pressure (ABP), and shown it feasible on 20 patients. We have also assembled and tested a device that allows collection of the data necessary to implement in real time, and to further refine this algorithm. Specific Aim #1: to test the feasibility of our approach to determining nICP on humans at several hospitals using invasive ABP (iABP), by collecting additional patient data to be used for refining the existing algorithm. Specific Aim #2: to perform the same feasibility tests and algorithm refinement with non-invasive ABP (nABP). Therefore, the work proposed here offers to perform steps necessary to bringing to space a non-invasive means of determining intracranial pressure. Key findings: in support of Aim #1, (a) In collaboration with our industrial partner and after completing the refinement of our prototype nICP device that also collects data necessary for finalizing the nICP algorithm, (b) we have used several of them to collect 91 useful data using that device and iABP thanks to its successful deployment at Harborview Medical Center (our Seattle hospital), from Wilford Hall Medical Center at Lackland Air Force Base in San Antonio, Texas, at Columbia-Presbyterian Medical Center in New York City, at Baylor Medical College in Houston Texas, at the University of Texas in Houston, Texas, and at the Barrow Neurological Institute, in Phoenix, AZ. In support of Aim #2, (a) we have identified three useful nABP devices; (b) we have shown that if their data quality is sufficient, that data from nABP will be sufficient to drive our nICP algorithm; (c) we have collected data from 6 patients using on nABP device at our Seattle hospital. Regrettably, two of the three devices have insufficient reproducibility relative to invasive arterial lines to be useful for our purpose. Furthermore, we have preliminary laboratory data on a complementary method for using ultrasound to determine nICP. Finally, with our industrial partner we have made technical advances toward the development of a 'user independent' transcranial Doppler system, which would allow the automated determination of nICP, and would also allow automated monitoring of cerebral blood flow and potential emboli into the brain within micro-g, high-g, and zero gravity environments. We are in contact with several interested parties within NASA regarding potential trials of these novel devices. Impacts: we spent the first year of this study laying the ground work for our proposed research, by establishing the medical sites, finishing our nICP prototype, and creating the necessary computational structures for data acquisition and processing. We have also demonstrated that we can collect useful data with our prototype system. These achievements allowed us to move forward towards achieving the goals of our research project, namely to collect and analyze sufficient patient data to refine and finalize our approach to nICP. Finally, we have begun the work necessary to eventually introduce into NASA an autonomous transcranial Doppler system. In the second year of this project we maintained our proposed pace of research. In particular, we deployed our data acquisition and prototype nICP device at the remote sites in collaboration with our industrial partner; we have used the web-based data transfer system to bring that data to our home institution for analysis on our computer cluster; we have tried out various nABP devices at our home institution within the context of our nICP research protocol; we have made further laboratory advances in the laboratory on a complementary means of non-invasively determining intracranial pressure; we have made, in collaboration with our industrial partner, further advances on the development of our user independent transcranial Doppler unit. It is our hope that this year we will lay the foundation for deploying a prototype user-independent TCD device within NASA's bed-rest test facilities. We anticipate continuing this pace of work in the third year, coming close to finalizing our algorithm for nICP and deploying for testing an autonomous TCD unit, in collaboration with our industrial partner. Our laboratory-based, and complementary approaches to predicting ICP non-invasively should enter animal testing (under separate funding by that time) during this year as well.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	1) non-invasively determining intracranial pressure (nICP): research in this area for terrestrial applications have been motivated by the vital role played by this medical variable in the management of individuals suffering from head trauma, subarachnoid hemorrhage and other sources of intracranial bleeding, as well as brain tumors, hydrocephalus, edema, among other disorders. We have demonstrated in our preliminary work for this proposal a novel means of determining nICP that has the best chance to date to capture this critical medical parameter whose current means of measurement requires that a neurosurgeon drill a hole through the skull. 2) We have begun the earliest steps towards automating the transcranial Doppler process for the purposes of deriving medically useful information from measurements of blood flow rate the middle cerebral artery. Given such a system, this would allow clinicians to easily and autonomously measure the rate of blood flow into the brain - a critical parameter in our approach to the non-invasive determination of intracranial pressure, as well as for the quantification of the existence and extent of emboli transport into the brain, and the quality of cerebral autoregulation. The monitoring of these measurements plays an important role in guiding the medical and/or surgical care of patients with a variety of neurological disorders, as listed above. 3) Others have noted in the literature that transcranial Doppler ultrasound, in conjunction with various pharmaceutical agents, can accelerate the destruction of unwanted blood clots in the brain - the major cause of stroke for several hundred thousand individuals in the USA every year. An autonomous transcranial Doppler system targetting the middle cerebral artery would allow the rapid and automatic focus of ultrasound onto the site of the majority of blood clots within the brain, making ultrasound-assisted thrombolysis readily deployable in the field.
Task Progress:	After its establishment, we have been using our unique the computational and logistical infrastructure necessary to most effectively pursue our human-based studies on the non-invasive determination of intracranial pressure. Regarding computational infrastructure, we have built a computer cluster of sufficient power and connectivity to efficiently run through what is approaching terabytes worth of human physiological data through our mathematical formalisms. Regarding logistical infrastructure, we have also actively used our web-based data entry system that allows all of our clinical sites to transmit the human data they collect in a HIPPA compliant way to the University of Washington for analysis. Our on-going modeling efforts for determining ICP based upon novel statistical measures remain consistent with our initial estimates of the complexity of the particular mathematical model we are using to relate in an empirical fashion readily observable human physiological data to intracranial pressure. We have deployed our nICP prototype device -

	useful at this stage for data acquisition and off-line analysis - at Harborview Medical Center in Seattle, where we have collected the bulk of human data that demonstrate the usefulness of the device. We have also successfully deployed it at the Barrow Neurological Institute and at Columbia University Hospital. This human physiological data also directly supports of our research effort effort toward the non-invasive determination of ICP. With these findings we are ready to collect the bulk of the human data we need to finalize our intracranial pressure algorithm by the end of the third year of this study, with algorithm finalization to occur in the fourth year.
Bibliography Type:	Description: (Last Updated: 11/21/2013)
Abstracts for Journals and Proceedings	Blizzard DJ, Vaswani PA, Kolve S, Ollos R, Kliot M, Mourad PD. "On selected methods for noninvasive determination of intracranial pressure." University of Washington's Annual Undergraduate Research Symposium, Seattle, WA, May 2007. University of Washington's Annual Undergraduate Research Symposium, Seattle, WA, May 2007. , May-2007
Abstracts for Journals and Proceedings	Blizzard DJ, Vaswani PA, Kolve S, Ollos R, Kliot M, Mourad PD. "Tapping the brain with ultrasound to noninvasively determine intracranial pressure." University of Washington's Annual Undergraduate Research Symposium, Seattle, WA, May 2007. University of Washington's Annual Undergraduate Research Symposium, Seattle, WA, May 2007. , May-2007
Abstracts for Journals and Proceedings	Mohr B, Mourad PD, Macaulay M, Ollos RJ, Newell DW, Sviri GE, Kliot M. "Noninvasive determination of intracranial pressure using ultrasound." Congress of Neurological Surgeons, San Diego, CA, Fall 2006. Proceedings of the Congress of Neurological Surgeons, Fall 2006. , Sep-2006
Abstracts for Journals and Proceedings	Vaswani PA, Ollos R, Mourad PD. "Noninvasive determination of intracranial pressure by means of vibroacoustography." University of Washington's Annual Undergraduate Research Symposium, Seattle, WA, May 2007. University of Washington's Annual Undergraduate Research Symposium, Seattle, WA, May 2007. , May-2007
Awards	Blizzard, Dan. "NASA Space Grant Summer Fellowship, June 2006." Jun-2006
Awards	Blizzard, Dan. "Research Fellowship for Advanced Undergraduates, March 2007." Mar-2007
Awards	Vaswani, Pavan. "NASA Space Grant Summer Fellowship, June 2006." Jun-2006
Awards	Vaswani, Pavan. "Research Fellowship for Advanced Undergraduates, March 2007." Mar-2007
Awards	Vaswani, Pavan. "The Barry M. Goldwater Scholarship (a nationally competed undergraduate fellowship), May 2007." May-2007
Patents	TBD. Filed, February 2005. Feb-2005 Mourad PD, Kliot M, Frederickson RCA. "Systems and methods for making noninvasive physiological assessments by detecting induced acoustic emissions."